Statistics Cheat Sheet (Ch.3–5)

Probability Basics

Random process ⇒ known outcomes, unknown result.

$$0 \le P(A) \le 1, \quad P(S) = 1$$

Disjoint: $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$. **General rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. **Complement:** $P(A^c) = 1 - P(A)$. **Independent:** $P(A \cap B) = P(A)P(B)$, $P(A|B) = P(A)$. **Conditional:** $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$. **At least one:** $1 - P(\text{none})$. Example: $P(\text{1 vegan in 3}) = 1 - (0.8)^3 = 0.488$.

Bayes' Theorem

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

Example: $P(D) = 0.01$, $P(+|D) = 0.95$, $P(+|D^c) = 0.05 \Rightarrow P(D|+) = \dfrac{0.95(0.01)}{0.95(0.01) + 0.05(0.99)} \approx 0.16$.

Discrete RVs

All probabilities ≥ 0 and sum to 1. $E[X] = \sum x_i P(x_i)$, $\text{Var}[X] = \sum (x_i - E[X])^2 P(x_i)$, $\sigma = \sqrt{\text{Var}[X]}$. **Linear combo:** $E[aX + bY] = aE[X] + bE[Y]$, $Var[aX + bY] = a^2 Var[X] + b^2 Var[Y]$ (if indep).

Binomial $X \sim Bin(n, p)$

$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $E[X] = np$, $Var[X] = np(1-p)$. **R:** `dbinom(k,n,p)`→$P(X = k)$; `pbinom(k,n,p)`→$P(X \le k)$; `pbinom(k-1,n,p,lower.tail=F)`→$P(X \ge k)$.
Ex: $P(X \ge 8)$ for n=10, p=0.3 `pbinom(7,10,0.3,lower.tail=F)`.

Poisson $X \sim Pois(\lambda)$

$P(X = k) = \dfrac{\lambda^k e^{-\lambda}}{k!}$, $E[X] = Var[X] = \lambda$. **R:** `dpois(k, )`→$P(X = k)$; `ppois(k, )`→$P(X \le k)$; `ppois(k-1, ,lower.tail=F)`→$P(X \ge k)$. *Ex:* λ=2, $P(X$=3)=`dpois(3,2)`=0.18.

Normal $X \sim N(\mu, \sigma)$

$Z = \dfrac{X - \mu}{\sigma} \Rightarrow Z \sim N(0,1)$. **R:** `pnorm(x, , )`→$P(X \le x)$; `pnorm(x, , ,lower.tail=F)`→$P(X > x)$; `qnorm(p, , )`→quantile. *Ex:* $P(X > 59.5)$ for $N(50, 5)$ `pnorm(59.5,50,5,lower.tail=F)` ≈ 0.029.
Normal Binomial: large $n \Rightarrow X \sim Bin(n,p) \approx N(np, \sqrt{np(1-p)})$, use ± 0.5 continuity correction.
Poisson Normal: large $\lambda \Rightarrow N(\lambda, \sqrt{\lambda})$.

Inference Overview

Point estimate: $\bar{x}$ or $\hat{p}$.
SEs: $SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$, $SE_{\bar{x}} = \frac{s}{\sqrt{n}}$.
95% CI: $\hat{p} \pm 1.96\, SE_{\hat{p}}$.
Z-test for proportion: $Z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$; reject H_0 if p-value < 0.05. Type I error = $\alpha = 0.05$.

Quick Memory

LLN: $\hat{p} \to p$ as $n \uparrow$. "At least 1" $\to 1 - P(\text{none})$. Continuity corr: ± 0.5. lower.tail=FALSE → right tail. Z 1.64 (90%), 1.96 (95%), 2.33 (98%). Right-skew mean > median. Left-skew mean < median. Indep check $\to P(A|B) = P(A)$. Binomial conds: fixed n, indep trials, const p. Poisson conds: rare indep events, rate λ. CLT sample means Normal. 95 % rule: mean ± 2 SD.
Mental anchors: Coin flip → Binomial; emails/hour → Poisson; heights → Normal.

General notes
- Proportion: $\hat{p} = x/n$ (successes / sample size).
- z^* = normal critical value (e.g. 1.96 for 95%).
- For tests: use pnorm on z to get p-values.

1. One-proportion CI & HT
CI for population proportion p:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Conditions: independent sample; at least 10 successes and 10 failures.
HT for $H_0 : p = p_0$:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

p-value from Z:
- $H_A : p < p_0$ (left): p-value = pnorm(Z)
- $H_A : p > p_0$ (right): p-value = 1 − pnorm(Z)
- $H_A : p \ne p_0$ (two-sided): p-value = $2\min\{\text{pnorm}(Z), 1 - \text{pnorm}(Z)\}$

R: prop.test (one proportion)

```
prop.test(x, n,
          p = p0,          # H0 value (optional)
          alternative = "two.sided", # or "less","greater"
          conf.level = 0.95,
          correct = FALSE)
```

Key:
- x: number of successes.
- n: sample size.
- p: p_0 in H_0 (if doing HT).
- alternative: direction of H_A.
- conf.level: e.g. 0.90, 0.95, 0.98, 0.99.
- correct = FALSE: no continuity correction (matches quiz solutions).

Output gives: CI for p, test statistic ($X^2 = Z^2$), p-value.

2. Two-proportion CI & HT
Two independent groups: (x_1, n_1), (x_2, n_2), proportions p_1, p_2.
Point estimates:

$$\hat{p}_1 = x_1/n_1, \quad \hat{p}_2 = x_2/n_2, \quad \hat{p}_1 - \hat{p}_2.$$

CI for $p_1 - p_2$:

$$SE_{\text{CI}} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* SE_{\text{CI}}$$

HT for $H_0 : p_1 - p_2 = \Delta_0$ (often $\Delta_0 = 0$):

$$\hat{p}_{\text{pool}} = \frac{x_1 + x_2}{n_1 + n_2}$$

$$SE_{\text{HT}} = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - \Delta_0}{SE_{\text{HT}}}$$

Then p-value as in one-proportion, according to H_A.
R: prop.test (two proportions)

```
prop.test(c(x1, x2), c(n1, n2),
          p = NULL,          # H0: p1 = p2 by default
          alternative = "two.sided", # or "less","greater"
          conf.level = 0.95,
          correct = FALSE)
```

Key:
- c(x1,x2): successes in each group.
- c(n1,n2): sample sizes.
- CI is for $(p_1 - p_2)$.
- Test is usually $H_0 : p_1 = p_2$ unless you specify p.

3. Chi-square tests
Goodness-of-fit (one categorical variable):

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}, \quad df = k - 1$$

Expected counts: $E_i = n \cdot p_i$. Reject H_0 if $\chi^2 > \chi^2_{1-\alpha,df}$.
Independence (two-way table $r \times c$):

$$E_{ij} = \frac{(\text{row total})_i (\text{col total})_j}{\text{grand total}}, \quad df = (r-1)(c-1).$$

R: chisq.test

```
# Goodness-of-fit
chisq.test(x = c(O1, O2, ...., Ok),
           p = c(p1, p2, ...., pk))
# Independence (2-way table)
tab <- matrix(c(...), nrow = r, byrow = TRUE)
chisq.test(tab)
```

Key:
- Output: χ^2 statistic, df, p-value.
- For GOF: p = vector of hypothesized probabilities.
- For independence: only give the table (no p).
- Reject if p-value $< \alpha$.

4. Margin of error & sample size
For a proportion CI:

$$ME = z^* \sqrt{\frac{p(1-p)}{n}}$$

Solve for n (desired ME):

$$n = \frac{z^2 p(1-p)}{ME^2}$$

If p unknown, use $p = 0.5$ (conservative, largest n).

R helpers (critical values):

```
qnorm(1 - alpha/2)   # z* for CI
qnorm(alpha)         # lower-tail critical z
qnorm(1 - alpha)     # upper-tail critical z
qchisq(1 - alpha, df)  # chi-square critical value
```